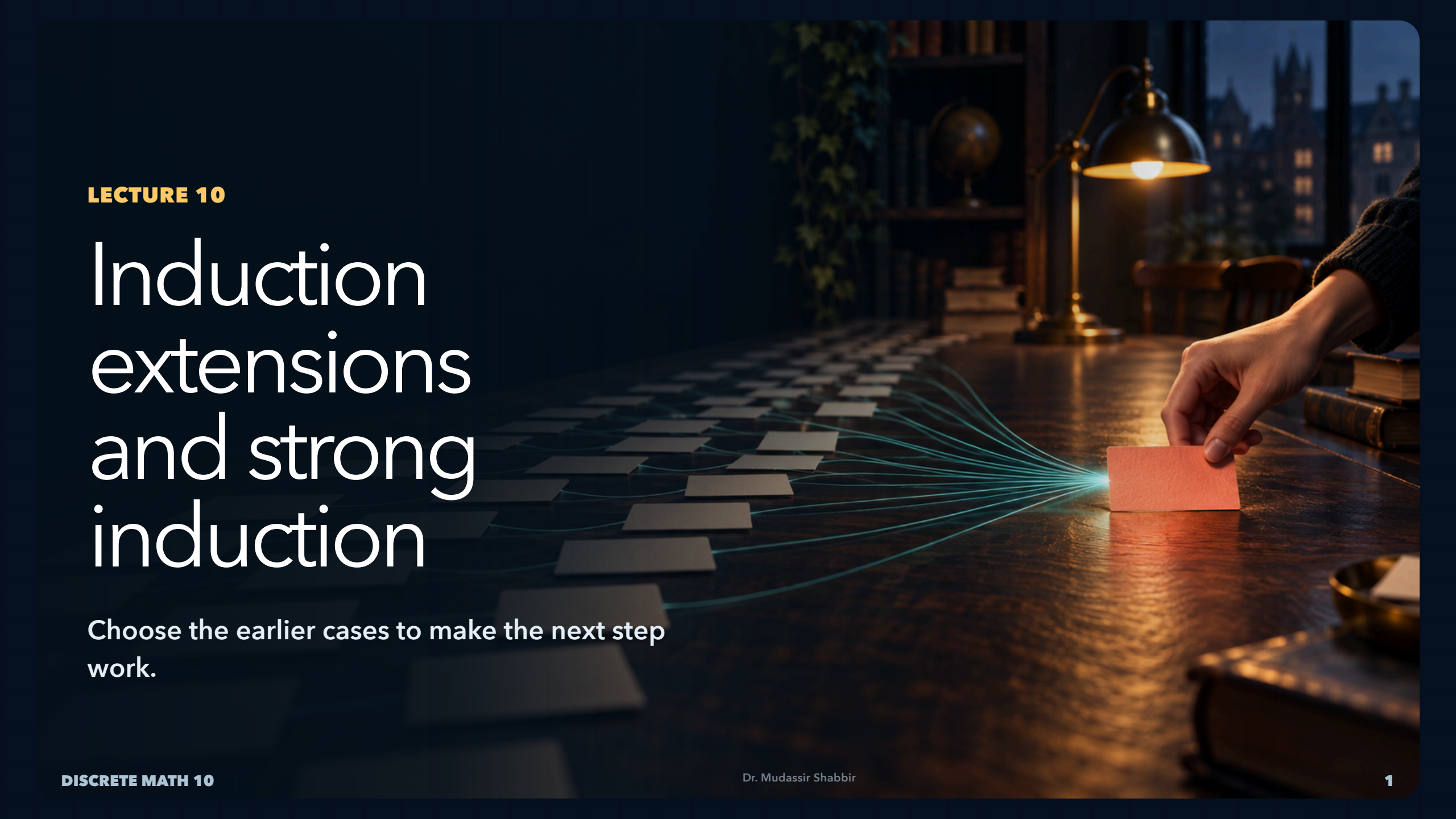


LECTURE 10

Induction extensions and strong induction

Choose the earlier cases to make the next step work.



WORKED EXAMPLE • PASCAL'S ROW SUMS • EXTENSION

Extension: each Pascal row doubles

Prove for every nonnegative integer n : the entries of row n sum to 2^n .

Base case

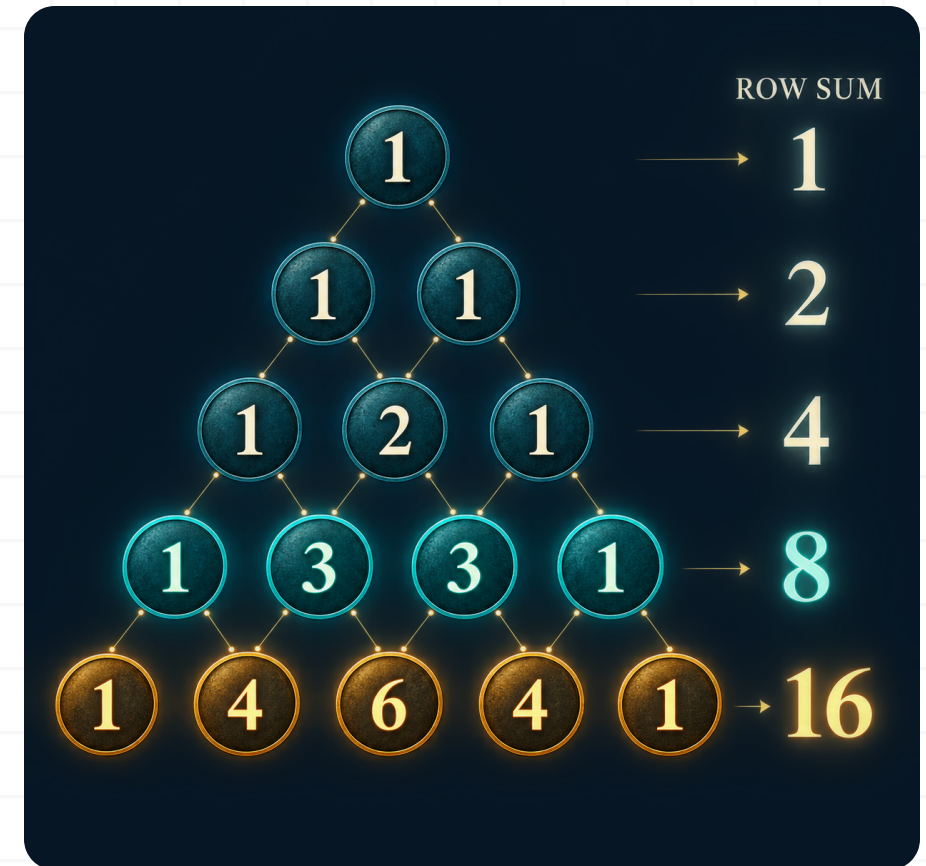
row 0 = {1}: $1 = 2^0$

True.

Inductive hypothesis

row k sums to 2^k

Assumed for arbitrary $k \geq 0$.



For example, row 3 is 1, 3, 3, 1: its sum is $8 = 2^3$.

Count each old entry twice

Pascal's rule

$$C(k+1, j) = C(k, j-1) + C(k, j)$$

Each interior entry adds the two above; entries beyond the row are zero.

Sum row $k+1$

$$= 2 \sum_j C(k, j) = 2 \cdot 2^k$$

Each entry of row k is counted twice.

Conclude

$$= 2^{k+1}$$

Exactly the claim at $n = k+1$. QED.

PART 2 • STRONG INDUCTION

Let the next case use every earlier case

The proof still advances one index at a time. Its hypothesis carries more memory.

$$P(n_0) \wedge \cdots \wedge P(k)$$

ORDINARY AND STRONG

The hypothesis is the only change

Ordinary induction

$$P(k)$$

Use the immediately previous case to prove $P(k + 1)$.

Strong induction

$$P(n_0), P(n_0 + 1), \dots, P(k)$$

Use any established earlier case to prove $P(k + 1)$.

The methods are logically equivalent; choose the hypothesis that fits the proof.

THE PROOF CONTRACT

Strong induction carries a complete prefix

To prove $P(n)$ for every integer $n \geq n_0$:

Base cases

$$P(n_0), \dots, P(n_0 + r)$$

Verify enough starting values for the planned step.

Strong hypothesis

$$P(i) \text{ holds for every } n_0 \leq i \leq k$$

Fix an arbitrary integer $k \geq n_0 + r$ and assume all these cases.

Inductive step

$$P(n_0) \wedge \dots \wedge P(k) \Rightarrow P(k + 1)$$

Use the earlier indices supplied by the structure.

The base window and the step must fit each other.

WORKED EXAMPLE • GUIDED PRIME FACTORIZATION

Every integer above 1 breaks into primes

Definition: A product may contain one or more prime factors.

Task: Prove that every integer $n \geq 2$ is a product of primes.

Base: 2 is a one-prime product. Assume every integer $2, \dots, k$ is a product of primes.

If $k + 1$ is prime, it is already done. Otherwise $k + 1 = ab$ with $2 \leq a, b < k + 1$, so $2 \leq a, b \leq k$.

Apply the hypothesis to both factors and join their prime products. By strong induction, every $n \geq 2$ is a product of primes.



First decide: what if the new number is prime? What if it is composite?

WORKED EXAMPLE • STRONG INDUCTION

Repair an illegal use of the hypothesis

A proof factors $k + 1 = ab$ and immediately applies the strong hypothesis to a and b . Why is that invalid if $a = 1$? State the bounds that make it valid.

If $a = 1$, then $b = k + 1$: the proof assumes the very case it is trying to prove.

Use a nontrivial factorization of a composite number: $a, b \geq 2$ and $a, b \leq (k + 1)/2 < k + 1$.

Hence $2 \leq a, b \leq k$.

Every factor used by the hypothesis must be smaller than the target.

WORKED EXAMPLE • STRONG INDUCTION

Build every postage total from 12

Definition: A total is buildable if it equals $4a + 5b$ for integers $a, b \geq 0$.

Task: Prove that every integer total $n \geq 12$ is buildable.

$$P(n) : n = 4a + 5b$$

Bases: $12 = 4 + 4 + 4$, $13 = 4 + 4 + 5$, $14 = 4 + 5 + 5$, and $15 = 5 + 5 + 5$.

Assume every total from 12 through k is buildable, where $k \geq 15$.

Then $k - 3$ is in that range. Build $k - 3$ and add one 4-cent stamp to obtain $k + 1$.

Four consecutive bases support a step that adds four.

WORKED EXAMPLE • STRONG INDUCTION

Why is one postage base not enough?

A proposed proof checks $P(12)$, then builds n by adding a 4-cent stamp to $n - 4$. Which totals are left unsupported? Repair the proof.

From 12 alone, this step reaches only 12, 16, 20, ... It does not establish 13, 14, 15.

Verify 12, 13, 14, 15. For every $n \geq 16$, the value $n - 4$ lies in the assumed range $12, \dots, n - 1$.

The base window must cover every chain created by the step.

WORKED EXAMPLE • STRONG INDUCTION

Use 3-cent and 5-cent stamps

Prove that every integer $n \geq 8$ can be written as $3a + 5b$ with integers $a, b \geq 0$. Choose the base cases to match a step that adds 3.

Bases: $8 = 3 + 5, 9 = 3 + 3 + 3, 10 = 5 + 5$.

For $k \geq 10$, assume every total $8, \dots, k$ is buildable. Since $8 \leq k - 2 \leq k$, build $k - 2$ and add 3 to get $k + 1$.

Strong induction proves every total from 8 onward is buildable.

Three consecutive bases support a step that adds three.